

Complex Analysis Homework #12

1. Suppose that $f : \mathbb{C} \rightarrow \hat{\mathbb{C}}$ is a meromorphic function with two periods ω_1, ω_2 , such that the ratio ω_1/ω_2 is real.
- (a) Suppose ω_1/ω_2 is rational. Show that the periodicity assumption is equivalent to being periodic with respect to some single ω' .
- (b) If ω_1/ω_2 is irrational, show f is constant.

1. Let us write $\omega_1/\omega_2 = p/q$ for some coprime integers p, q . Let us define $\omega_0 := \omega_1/p$. We need to show that (1) f is periodic with respect to ω_0 and (2) that ω_1, ω_2 are multiples of ω_0 .

We show (2) first. Notice that $\omega_1 = p\omega_0$ by definition. And also, $\omega_2 = \frac{q\omega_1}{p} = q\omega_0$.

Next we show (1), that f is periodic with respect to ω_0 . Recall Bézout's identity gives us that there exists integers m, n such that $mp + nq = 1$ since p, q are coprime. Rearranging gives us that

$$\begin{aligned} \frac{\omega_1}{p} &= \omega_1 \left(m + n \frac{q}{p} \right) \\ &= \omega_1 \left(m + n \frac{\omega_2}{\omega_1} \right) \\ \omega_0 &= m\omega_1 + n\omega_2. \end{aligned}$$

Since f is periodic in ω_1 and ω_2 it thus follows that f is periodic in ω_0 .

2. Let $\tau := \omega_1/\omega_2$, we have that τ is irrational. Let $m, n \in \mathbb{Z}$ and notice that $m\omega_1 + n\omega_2 = \omega_2(m\tau + n)$. Now since τ is irrational we have that the set $\{\omega_2(m\tau + n)\}$ is dense in the line $\mathbb{R}\omega_2$. We will show that f is thus constant on the line $\mathbb{R}\omega_2$.

Let $c \in \mathbb{R}\omega_2$ be a point where f is holomorphic. Since $\{\omega_2(m\tau + n)\}$ is dense in $\mathbb{R}\omega_2$ we have a sequence $z_k = \omega_2(m_k\tau + n_k)$ with $m_k, n_k \in \mathbb{Z}$ which converges to c . Since f is holomorphic at c we have that it is in particular continuous at c , and so $f(c) = \lim_{k \rightarrow \infty} f(z_k)$. Moreover since $\omega_2(m_k\tau + n_k) = m_k\omega_1 + n_k\omega_2$ for all k we have that $f(\omega_2(m\tau + n)) = f(0)$ for all $m, n \in \mathbb{Z}$. But then $f(c) = \lim_{k \rightarrow \infty} f(z_k) = f(0)$, i.e. f is constant along $\mathbb{R}\omega_2$.

We have shown that f is constant on a closed subset of its domain of holomorphicity. It follows by the identity theorem that f is thus constant on its entire domain.

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2. Given a lattice $\Lambda \subset \mathbb{C}$, addition of complex numbers naturally descends to a group law on the torus $\mathbb{C}/\Lambda$. Let $F : \mathbb{C}/\Lambda \rightarrow \hat{\mathbb{C}}$ be a meromorphic function with zeros at $z_1, \dots, z_n$ and poles at $p_1, \dots, p_m$ (appearing with multiplicity equal to the order). Prove that

$$\sum z_i - \sum p_j = 0,$$

where the addition is with respect to the group law described above.

Since F is a meromorphic function on $\mathbb{C}/\Lambda$ it lifts to a doubly periodic meromorphic function on $\mathbb{C}$, let us also refer to this meromorphic function on $\mathbb{C}$ as F (I will note when we start thinking about it as a function on the torus again). Consider a fundamental parallelogram P of Λ in $\mathbb{C}$ such that F has no zeroes or poles on the boundary of P . Recall that the following variation of the argument principle integral counts the sum of the points which gives zeroes and poles for F .

$$\int_{\partial P} z \cdot \frac{f'(z)}{f(z)} dz = \sum_{z_i \in \mathbb{C}} z_i - \sum_{p_j \in \mathbb{C}} p_j.$$

Let us normalize our lattice so that one of the generators is given by 1 and the other is given by $\tau \in \mathbb{H}$. Let us label the sides of our fundamental parallelogram by $\alpha, \alpha', \beta, \beta'$. Then notice that we have the following identity

$$\begin{aligned} \int_{\beta} z \cdot \frac{f'(z)}{f(z)} dz &= \int_{-\beta' + \tau} z \cdot \frac{f'(z)}{f(z)} dz \\ &= \int_{-\beta'} (z + \tau) \cdot \frac{f'(z + \tau)}{f(z + \tau)} dz \\ &= \int_{-\beta'} (z + \tau) \cdot \frac{f'(z)}{f(z)} dz \\ &= - \int_{\beta'} (z + \tau) \cdot \frac{f'(z)}{f(z)} dz, \end{aligned}$$

where the third equality follows since f is periodic in τ and so f' is also periodic in τ . Similar reasoning gives that

$$\int_{\alpha} z \cdot \frac{f'(z)}{f(z)} dz = - \int_{\alpha'} (z + 1) \cdot \frac{f'(z)}{f(z)} dz.$$

Now, consider the following

$$\begin{aligned}\int_{\partial P} z \cdot \frac{f'(z)}{f(z)} dz &= \int_{\alpha+\beta+\alpha'+\beta'} z \cdot \frac{f'(z)}{f(z)} dz \\ &= \int_{\alpha'} \frac{f'(z)}{f(z)} dz - \tau \int_{\beta'} \frac{f'(z)}{f(z)} dz.\end{aligned}$$

Now, we claim (as prompted by the book hint) that the integral $\int f'(z)/f(z) dz$ along any of the side lengths of the fundamental parallelogram is some multiple of $2\pi i$. Consider

$$\int_{\alpha'} \frac{f'(z)}{f(z)} dz = \int_{z_0}^{z_0+\tau} \frac{f'(z)}{f(z)} dz = \log(f(z_0)) - \log(f(z_0 + \tau)),$$

where z_0 is one endpoint for α' . Note that, since we are integrating just over an interval in $\mathbb{C}$, we can take the branch cut of $\log$ to be away from either of the line segments that we care about. Now recall that f is periodic with respect to τ , i.e., $f(z_0) = f(z_0 + \tau)$. That means, the image of f under α' traces out some closed curve in $\mathbb{C}^\times$ (recall that we chose P so that f does not vanish on ∂P). It follows then that

$$\int \frac{f'(z)}{f(z)} dz = k(2\pi i),$$

for some integer k .

Now returning to the integral that we are concerned with, we thus have

$$\int_{\alpha'} \frac{f'(z)}{f(z)} dz - \tau \int_{\beta'} \frac{f'(z)}{f(z)} dz = (2\pi i)(k_1 - k_2\tau).$$

For some $k_1, k_2 \in \mathbb{Z}$. Now notice that $k_1 - k_2\tau$ is a lattice point. And so $k_1 - k_2\tau \equiv 0 \pmod{\Lambda}$. In other words, we have shown

$$\sum z_i - \sum p_j = \int_{\partial P} z \cdot \frac{f'(z)}{f(z)} dz \equiv 0 \pmod{\Lambda}.$$

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3. Given a lattice $\Lambda \subset \mathbb{C}$, show that the series

$$\sum_{\omega \in \Lambda^*} \frac{1}{|\omega|^2}$$

is divergent.

Recall that without loss of generality we can take our lattice Λ to be generated by 1 and $\tau = a + bi$ with $a, b \in \mathbb{R}$ and $b > 0$. We can also choose generators of our lattice so that $a > 0$. With this notation we have that $\omega = m + n\tau$ for integers m, n , and so

$$|\omega|^2 = |m + n(a + bi)|^2 = (m + na)^2 + (nb)^2.$$

Our series of interest becomes

$$\sum_{\omega \in \Lambda^*} \frac{1}{|\omega|^2} = \sum_{m, n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2}.$$

For m, n not both simultaneously zero. Consider this sum for some fixed n . We can split our sum

$$\begin{aligned} \sum_{m, n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2} &= \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} + \sum_{(m+na)^2 > (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} \\ &\geq \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} \\ &\geq \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{2(nb)^2} \\ &= K(n) \cdot \frac{1}{2(nb)^2}. \end{aligned}$$

Where $K(n)$ is the number of integers m satisfying $(m + na)^2 \leq (nb)^2$. The second line follows since the latter sum on the first line contains only positive terms. The third line follows since, by the condition in the sum, $(nb)^2 \geq (m + na)^2$ for each term. Counting the number of integers m satisfying $(m + na)^2 \leq (nb)^2$ gives us that $K(n) \sim 2|nb|$ and so we have

$$\sum_{m, n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2} \geq \frac{1}{|nb|}.$$

Now we can show that our series of interest diverges

$$\sum_{\omega \in \lambda} \frac{1}{|\omega|^2} = \sum_{m,n \in \mathbb{Z}} \frac{1}{(m+na)^2 + (nb)^2} \geq \sum_{n \in \mathbb{Z}} \frac{1}{|nb|} = \frac{1}{b} \sum_{n \in \mathbb{Z}} \frac{1}{|n|},$$

which diverges.

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4. Let S be the strip $\{z : \operatorname{Im}(z) > 0, -\pi/2 < \operatorname{Re}(z) < \pi/2\}$, and let $f : \mathbb{H} \rightarrow S$ be the Riemann map whose extension to the boundary of $\mathbb{H}$ takes $(-1, 1, \infty)$ to $(-\pi/2, \pi/2, \infty)$. Show that f^{-1} can be extended to a map $\mathbb{C} \rightarrow \mathbb{C}$, and prove that it coincides with a familiar function.

We approach this problem by attempting to construct the desired Riemann map and its inverse.

First we construct a candidate Riemann map $\mathbb{H} \rightarrow S$ with the prescribed images. As a first approach, let $f : \mathbb{H} \rightarrow S$ via $f(z) = -i \log(iz)$, with the branch cut in the lower right quadrant of $\mathbb{C}$.

Consider $f : \mathbb{H} \rightarrow \mathbb{C}$ via $f(z) = -i \log(iz)$ with the branch cut in the lower half plane. Notice $f(1) = -i \log(i) = -i \log(e^{i\pi/2}) = -i(i\pi/2) = \pi/2$. Likewise, a similar calculation will give that $f(-1) = -\pi/2$ and $f(\infty) = \infty$. Since we chose the branch cut appropriately, we have that f is holomorphic on $\mathbb{H}$. Consider the image of f . Let $z = re^{i\theta} \in \mathbb{H}$ so that $r > 0$ and $0 < \theta < \pi$. Now

$$\begin{aligned} f(z) &= -i \log(ire^{i\theta}) \\ &= -i \log(re^{i(\theta+\pi/2)}) \\ &= -i(\ln(r) + i(\theta + \pi/2)) \\ &= (\theta + \pi/2) - i \ln(r). \end{aligned}$$

We have that the image has real part contained in $-\pi/2$ and $\pi/2$, but the imaginary part takes any value. And so the image of this map is the non-truncated strip bounded just by the lines $z = -\pi/2$ and $z = \pi/2$. Although, f has inverse given by $w \mapsto e^w$.

This calculation may give us an insight the following map, as a second guess. Consider $g(z) : S \rightarrow \mathbb{H}$ given by $g(z) = \sin(z)$. This is holomorphic on S and extends continuously to the boundary since it is entire as a function $\mathbb{C} \rightarrow \mathbb{C}$. We need to verify that g is a biholomorphism to $\mathbb{H}$ and that it takes the prescribed triple to the prescribed image.

First, we need to verify that the image of g is $\mathbb{H}$. Write $z = a + bi$ with $-\pi/2 < a < \pi/2$

and $b > 0$. Now consider the following

$$\sin(a + bi) = \sin(a) \cos(bi) + \cos(a) \sin(bi).$$

Recall the identities

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i} \quad \cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

In particular

$$\sin(ib) = \frac{e^{-b} - e^b}{2i} \quad \cos(ib) = \frac{e^{-b} + e^b}{2} \in \mathbb{R}.$$

And so we have that the real part of $\sin(a + bi)$ is given entirely by the second term, $\operatorname{Im}(\sin(a + bi)) = \operatorname{Im}(\cos(a) \sin(bi))$. Consider

$$\begin{aligned} \cos(a) \sin(bi) &= \cos(a) \frac{e^{-b} - e^b}{2i} \\ &= \cos(a) (-2i) \frac{e^{-b} - e^b}{4} > 0. \end{aligned}$$

Where the final inequality follows since $-\pi/2 < a < \pi/2$ gives $\cos(a) > 0$, and $b > 0$ gives $e^{-b} - e^b < 0$. That is

$$\operatorname{Im}(\sin(a + bi)) > 0$$

And so the image of g is contained in $\mathbb{H}$.

Consider the prescribed triples. We have $\sin(-\pi/2) = -1$ and $\sin(\pi/2) = 1$. Note that $\sin$ does not have a well-defined extension to infinity, when considered as a function $\mathbb{C} \rightarrow \mathbb{C}$. In particular, if we restrict our attention to the real line, $\sin(x)$ does not grow unboundedly as $x \rightarrow \infty$ along $\mathbb{R}$ (the more formal way to argue this would be to study what happens with $\lim_{z \rightarrow 0} \sin(1/z)$). However, since we are considering $\sin$ as a function on S , the only way for z to approach ∞ is by sequences z_n where $-\pi/2 < \operatorname{Re}(z_n) < \pi/2$ and $\operatorname{Im}(z_n) > 0$ for all n . In other words, we can only construct sequences that approach infinity by following the positive imaginary direction. For all such sequences $|\sin(z_n)|$ does indeed grow without bound (in particular $(e^{-b} - e^b)/4 \rightarrow -\infty$ as $b \rightarrow \infty$). Thus we have that $\sin(\infty) = \infty$. We have shown that g takes $(-\pi/2, \pi/2, \infty) \mapsto (-1, 1, \infty)$.

Lastly, we argue that g is a bijection. Notice that the quantity $(e^{-b} - e^b)/4$ is surjective to $\mathbb{R}_-$ for $b > 0$, and so $\operatorname{Im}(g)$ takes every positive imaginary value. Likewise $(e^{-b} + e^b)/2$

is surjective to $\mathbb{R}_+$ for $b > 0$, moreover, $\sin(a)$ takes each value between $-1, 1$ as $-\pi/2 < a < \pi/2$. And so $\operatorname{Re}(g)$ takes every real value. We can thus achieve any value in the upper half plane via g by tuning a, b .

We argue that g is injective. Suppose we had $z_1, z_2 \in S$ such that $g(z_1) = g(z_2)$. Recalling the exponential identity for $\sin$ this becomes the condition

$$k - \frac{1}{k} = m - \frac{1}{m},$$

where $k = e^{iz_1}$ and $m = e^{iz_2}$. This can be rearranged as follows

$$\begin{aligned} (k - m) - \left(\frac{1}{k} - \frac{1}{m} \right) &= 0 \\ (k - m) - \left(\frac{m - k}{mk} \right) &= 0 \\ (k - m) \left(1 + \frac{1}{mk} \right) &= 0. \end{aligned}$$

Now if $k = m$ then we have $e^{iz_1} = e^{iz_2}$ which implies $iz_1 = iz_2 + 2\pi i\ell$ for some integer ℓ . However this then gives $z_1 = z_2 + 2\pi k$, and no two elements of S can differ by a nonzero integer multiple of $2\pi i$ since the strip has real width π . In other words $k = m$ implies $z_1 = z_2$. Now, on the other hand, consider the condition $1 + 1/mk = 0$. We can rearrange this to acquire $mk = -1$. First, note that m, k are contained in the right half plane. Indeed if $a + bi$ satisfies $-\pi/2 < a < \pi/2$ then $e^{i(a+bi)} = e^{ia-b} = e^{-b}e^{ia}$ has argument $-\pi/2 < a < \pi/2$. There exist no m, k in the right half plane which satisfy $mk = -1$; there are no two angles in $[-\pi/2, \pi/2]$ that we can sum to π . And so the condition $1 + 1/mk = 0$ gives a contradiction. We have shown that g is injective as a map $S \rightarrow \mathbb{H}$. We have overall shown that g is a biholomorphism $S \rightarrow \mathbb{H}$.

We have shown that $g(z) = \sin(z)$ is the inverse for the claimed Riemann map. And indeed this map can be extended to a holomorphic map on all of $\mathbb{C}$ since $\sin(z)$ is entire.

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